Modeling Expected Health of Different Ages Around the World in Relation to Environmental Factors

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Outline

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2. Age-structured population model
3. Environmental and health related stressors
4. Parameterization and simulation setup
5. Baseline simulation results
6. Projected scenario analysis
7. Key findings and future work

Environmental stressors and long term population health

- Population health is increasingly shaped by long term environmental instability rather than isolated events.
- Factors such as food insecurity, air quality, climate driven droughts, and global disease outbreaks disproportionately affect vulnerable age groups.
- Understanding how these stressors interact with demographic structure is essential for long term health and population planning.

Study scope and objectives

- Simulations are conducted over a long term horizon (65 years) for six countries: Canada, United States, Australia, United Kingdom, Japan, and Brazil.
- The primary objective is to evaluate how persistent environmental stressors and discrete shock events influence age specific population dynamics.

Modeling framework

- The population is divided into four age-structured compartments: children (0–14), youth (15–29), adults (30–64), and seniors (65+).
- Individuals transition between age groups through aging, and enter the system through births.
- Each age compartment experiences age-specific mortality influenced by baseline rates and external stressors.
- The model tracks long-term changes in population composition.

Environmental stressors and scenario design

- Chronic environmental stressors (food insecurity and air quality) act continuously by increasing age-specific mortality rates.
- Discrete shock events (droughts and disease outbreaks) occur stochastically and temporarily amplify mortality.
- The model is simulated over a 65-year horizon to capture cumulative and compounding effects.
- Scenario-based simulations compare baseline conditions against intensified stressor and shock environments.

Governing equations: age-structured population model

$$\frac{dP_k}{dt} = b_y P_y(t) + b_a P_a(t) - \Big(d_k + f_k F + A_k A\Big) P_k(t) - \frac{P_k(t)}{15}, \tag{1}$$

$$\frac{dP_y}{dt} = \frac{P_k(t)}{15} - \Big(d_y + f_y F + A_y A\Big) P_y(t) - \frac{P_y(t)}{15}, \tag{2}$$

$$\frac{dP_a}{dt} = \frac{P_y(t)}{15} - \Big(d_a + f_a F + A_a A\Big) P_a(t) - \frac{P_a(t)}{35}, \tag{3}$$

$$\frac{dP_s}{dt} = \frac{P_a(t)}{35} - \Big(d_s + f_s F + A_s A\Big) P_s(t). \tag{4}$$

- P_k, P_y, P_a, P_s denote the populations aged 0–14, 15–29, 30–64, and 65+, respectively.
- Births enter the children compartment through age specific fertility rates b_y and b_a.
- Mortality includes baseline death rates d_i and chronic disease contributions from food (F) and air quality (A).

Model parameters: fixed vs. estimated

- **Fixed parameters (data-driven):**
  - Age specific birth rates, baseline death rates, and aging rates.
  - Chronic disease proportions for food and air-related mortality.
  - Initial population sizes for each age group.
- **Estimated / scenario parameters:**
  - Chronic disease mortality multipliers for food (F) and air quality (A).
  - Drought and disease shock multipliers by age group.

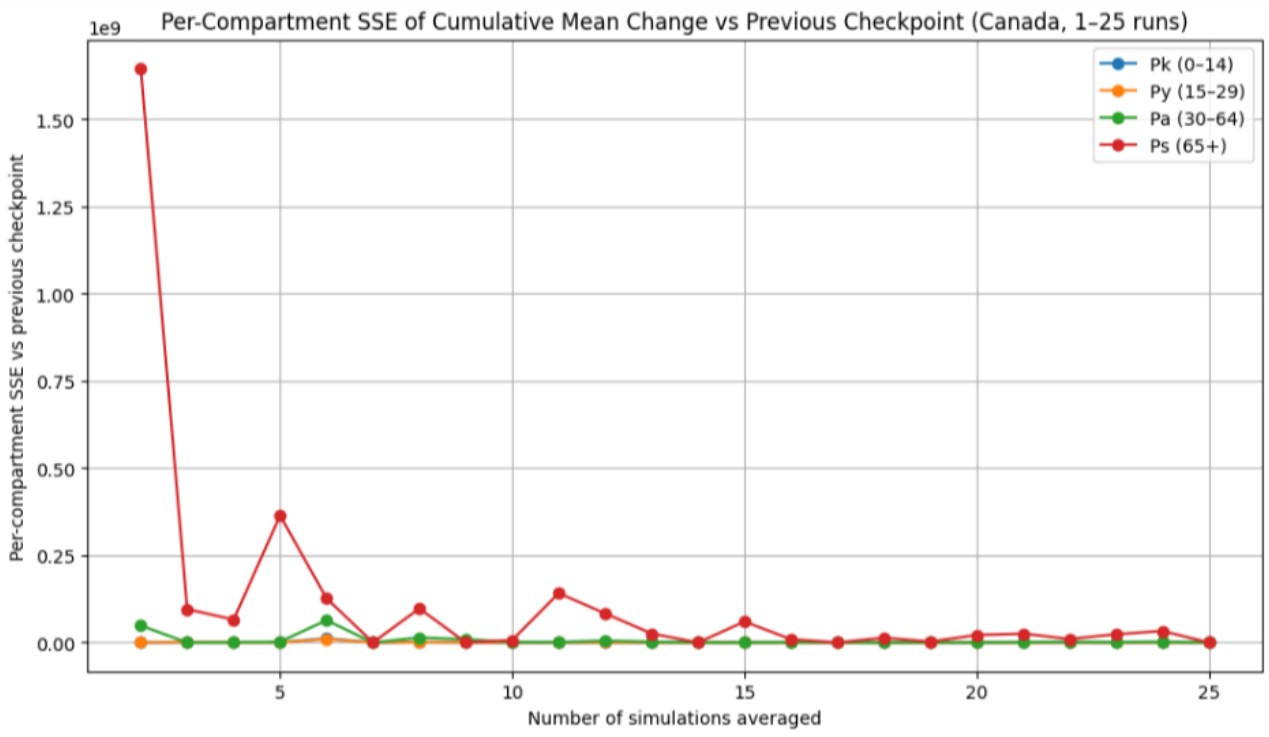
Model assumptions

- Diabetes and asthma mortality rates are identical across countries, and risk is uniform within each age compartment.
- Aging occurs uniformly within each compartment, and births occur only from the youth and adult groups.
- Birth and baseline death rates remain constant.
- Drought and disease events are modeled as independent Bernoulli processes with fixed annual probabilities.
- Discrete disease shocks apply age specific mortality patterns randomly selected from historical pandemics and act multiplicatively on baseline mortality.

Simulation setup

- Simulations are run over a 65 year horizon with a yearly time step.
- Model parameters are updated dynamically at each time step to reflect active drought or disease events.
- 20 simulation runs are performed to capture variability introduced by stochastic events.

Convergence of stochastic simulations

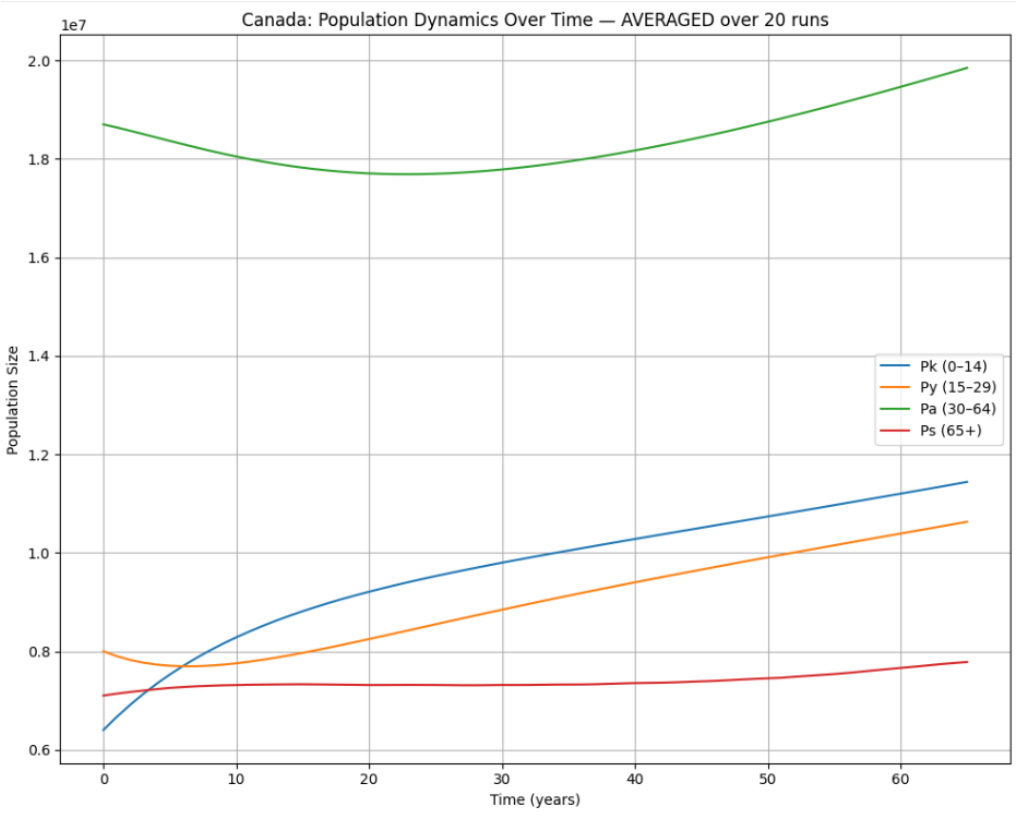


- The cumulative mean stabilizes by ≈ 20 runs (SSE of mean change becomes negligible across compartments).

Simulation scenarios

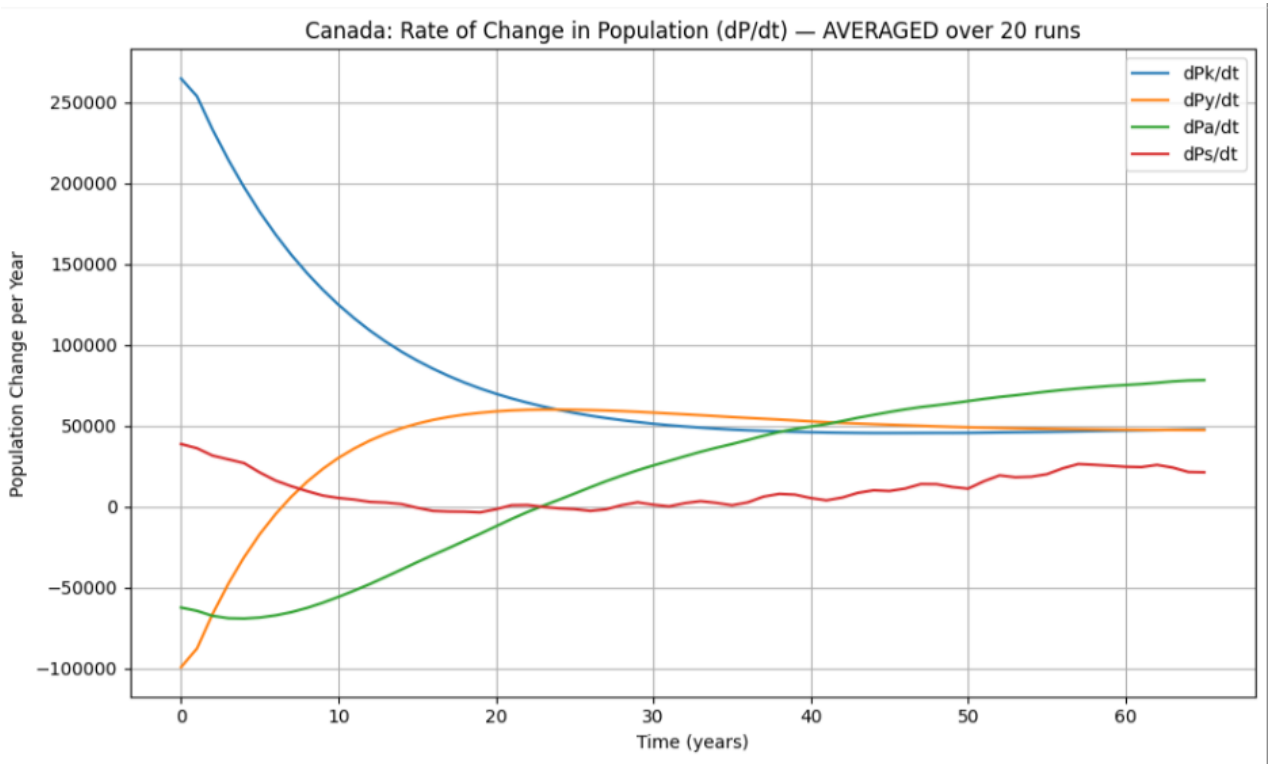
- **Baseline scenario:**
  - Constant demographic parameters and chronic disease effects.
  - Stochastic drought and disease events included at baseline probabilities.
- **Scenario 1: Time-varying birth and death rates**
  - Birth and death rates evolve over time based on projected demographic trends.
- **Scenario 2: Elevated chronic disease burden**
  - Increased food and air-related mortality to reflect higher chronic disease prevalence.
- **Scenario 3: Increased environmental shock frequency**
  - Higher frequency of stochastic drought and disease events.

Baseline results: Canada



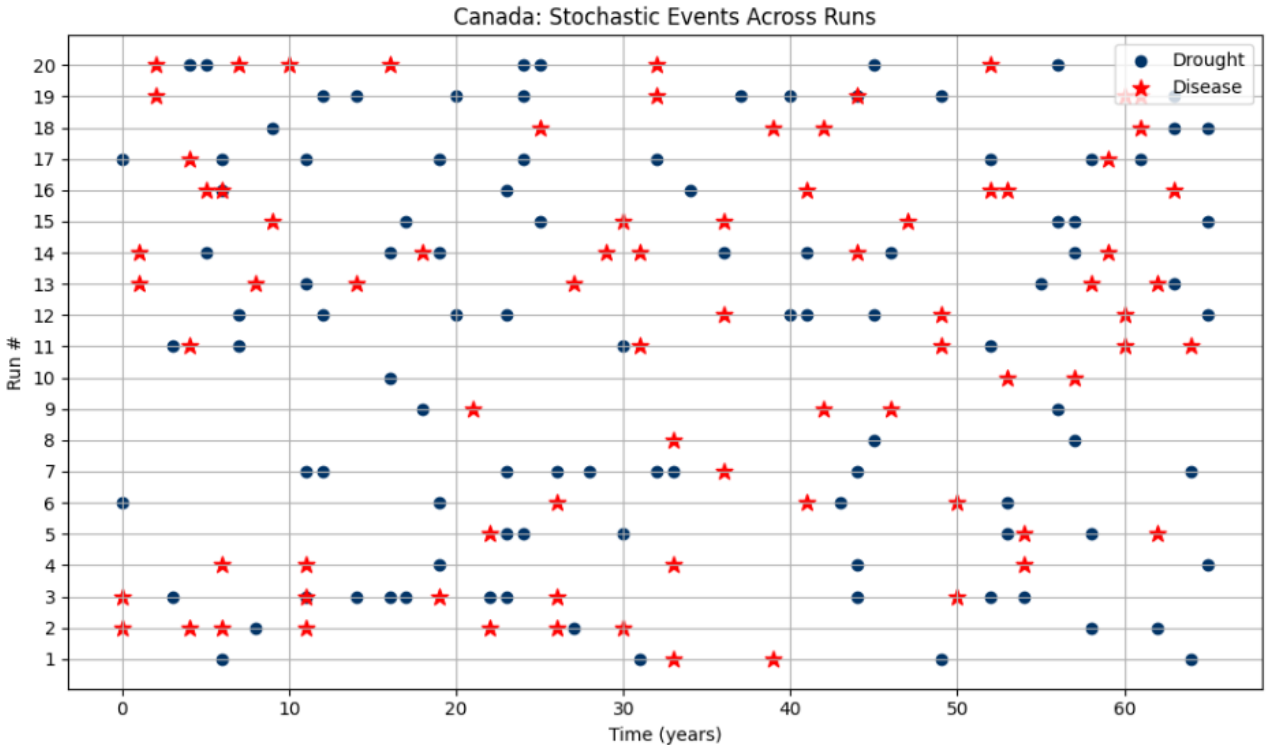
- Under baseline conditions, Canada exhibits gradual population aging, with slower growth in younger cohorts and sustained growth in older age groups.

Baseline results: Canada



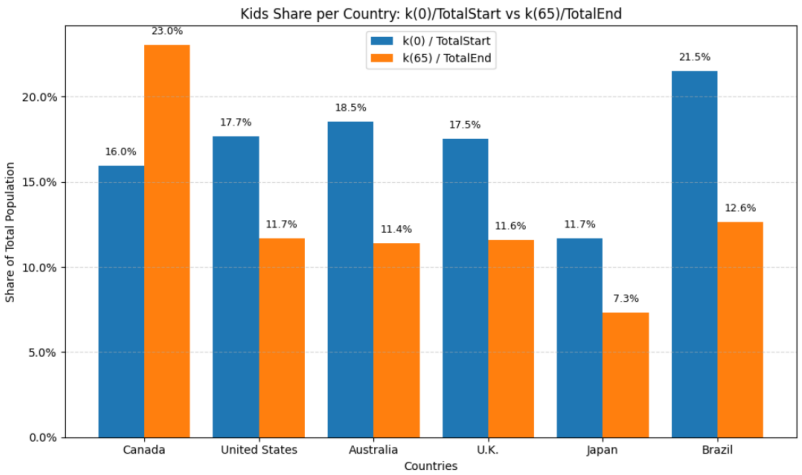
- Rates of change stabilize over time.

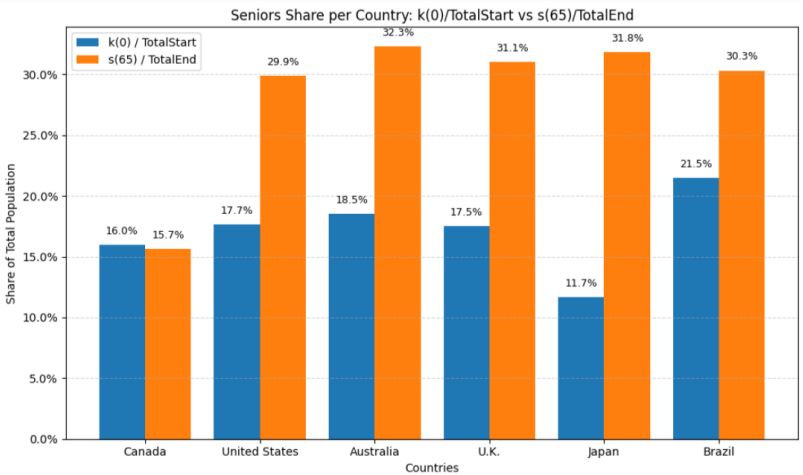
Baseline results: Canada



- Drought and disease shocks occur intermittently and independently across time.

Baseline results



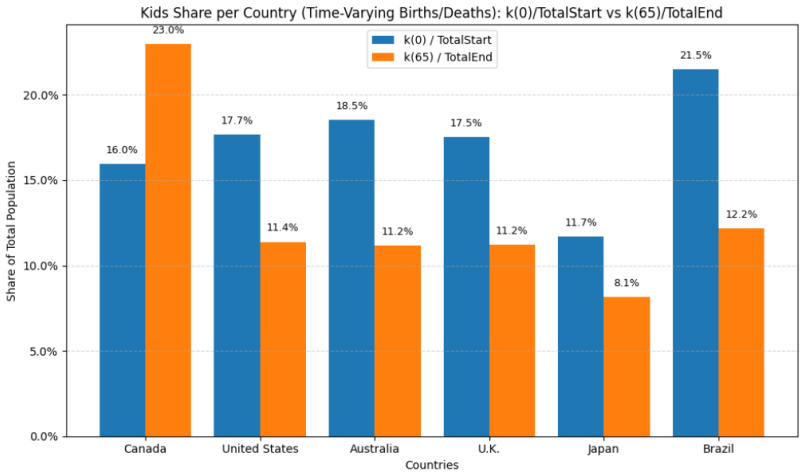


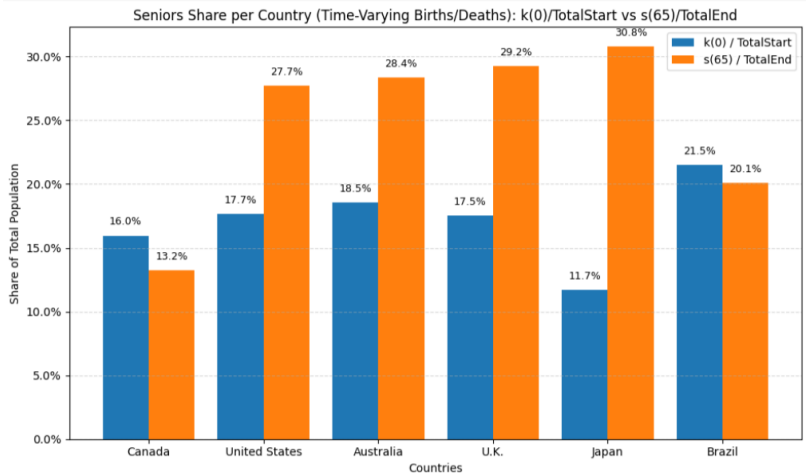
- Baseline age distributions vary substantially across countries, particularly in child and senior population shares.
- These differences provide important context for interpreting scenario-based outcomes.

Scenario 1: Time-varying birth and death rates

- Birth and death rates evolve over time based on projected national demographic trends.
- All other model components remain unchanged from the baseline scenario.

Scenario 1 results: Demographic projections



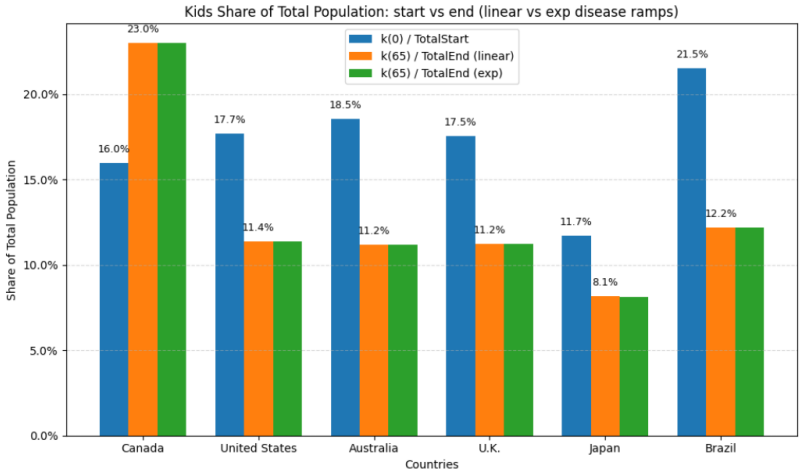


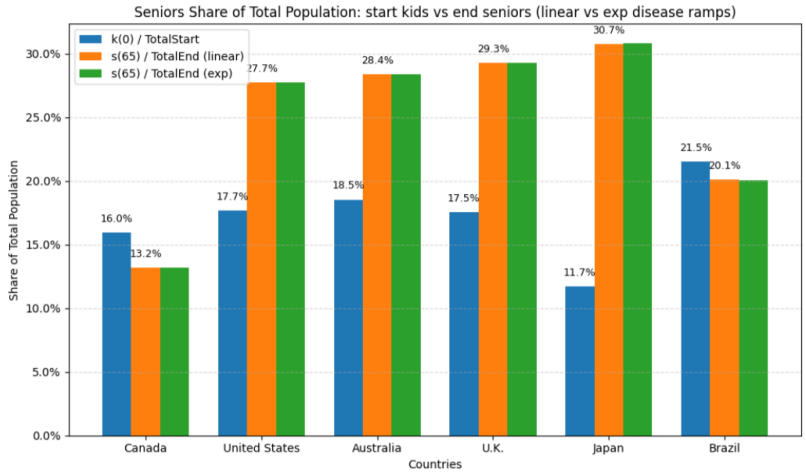
- Projected birth/death rates shift age composition relative to baseline, amplifying aging trends in several countries.
- The largest changes appear in the child and senior shares, which motivates comparing scenario impacts across countries.

Scenario 2: Elevated chronic disease burden

- Food and air-related mortality rates are increased across all age groups.
- Demographic parameters and shock probabilities remain fixed.
- This scenario examines sustained health deterioration rather than abrupt shocks.

Scenario 2 results: Chronic disease effects



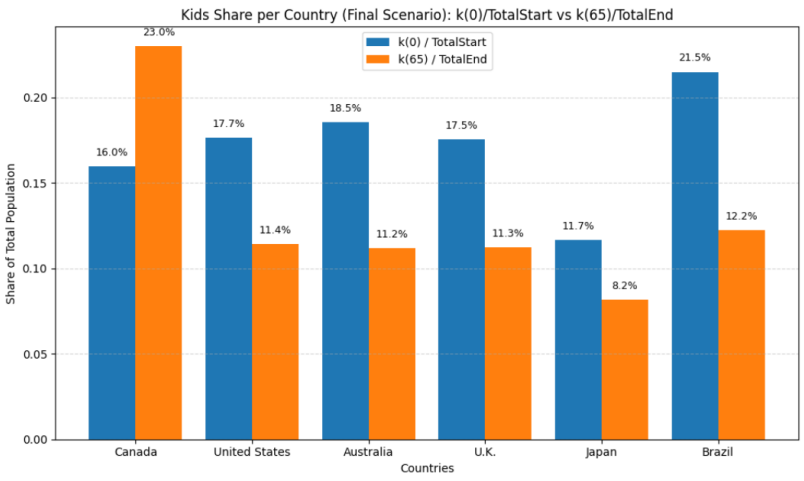


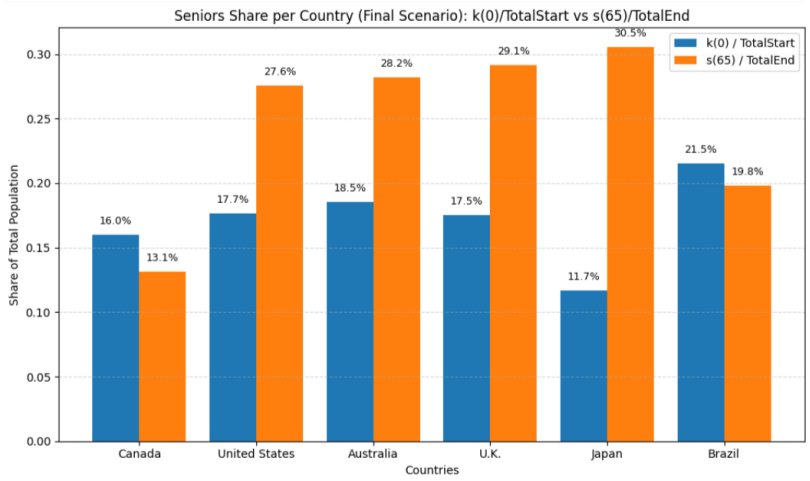
- Increasing chronic disease mortality produces a measurable shift in age structure relative to baseline.
- Impacts differ by country, reflecting differences in starting age distributions and baseline vulnerability.

Scenario 3: Increased environmental shock frequency

- The frequency of drought and disease shock events is increased.
- Shock severity and age-specific mortality patterns remain unchanged.

Scenario 3 results: Environmental instability





- More frequent shocks create larger deviations from baseline age composition, especially in the most vulnerable compartments.

Key findings across scenarios

- Long-term outcomes are strongly driven by initial age structure, with countries starting older exhibiting faster population decline under all scenarios.
- Time varying birth and death rates substantially accelerate aging, particularly through sustained reductions in child cohorts and slower replenishment of the working-age population.
- Elevated chronic disease burden produces compounding population loss over time, with disproportionate impacts on youth and working age groups due to sustained mortality pressure.
- Increased frequency of environmental shock events limits recovery between shocks, leading to greater variability and reduced long run demographic stability.
- Cross country differences in vulnerability persist across all scenarios, emphasizing that identical stressors can produce markedly different outcomes depending on demographic context.

Future work

- Expand and validate stressor parameterization with richer datasets.
- Add more countries/regions and explore inequality within countries.
- Refine health definition/indices and incorporate additional stressor feedbacks.

Thank you for listening!